EXECUTIVE SUMMARY

Customer Churn

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Marketing analysis has identified a significant efficiency gap within the company’s cumulative media investment. By analyzing 120 weeks of historical data, I developed a predictive model that explains **78% of sales fluctuations.**

The primary value of this project is the transition from static reporting to **Dynamic Decision Support** via the deployed Marketing Mix Modeling Dashboard. This tool allows leadership to move from hindsight to active foresight, providing a high-confidence environment to identify "wasted spend" in low-impact channels and recapture revenue that was previously unquantified.

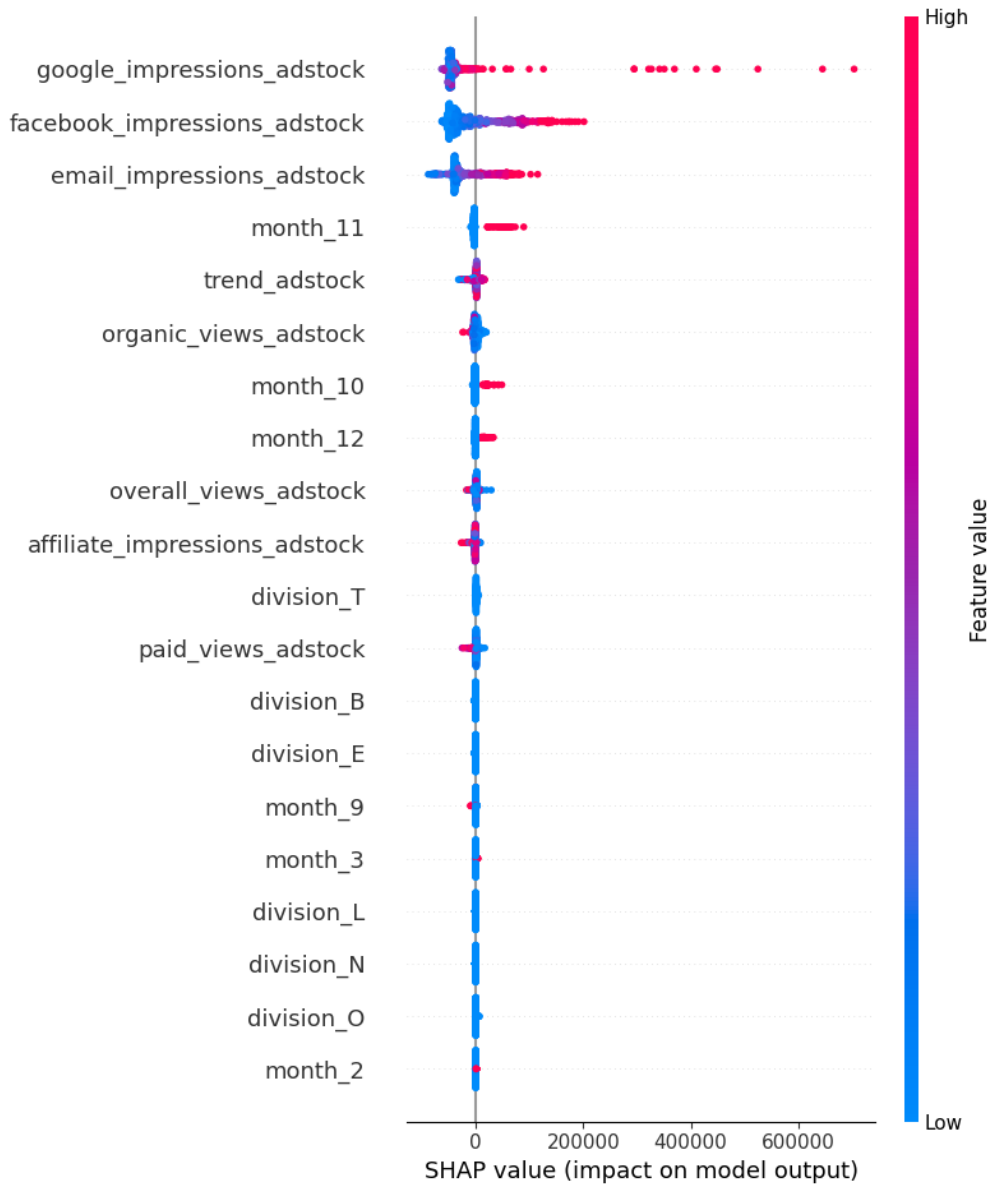
Strategic Diagnosis: What Drives Revenue?

Data analysis revealed that current budget performance is driven by two specific structural factors:

- **The Adstock Gap:**
  - **High-Impact Anchors**: **Affiliate Impressions α = 0.95)** and **Organic Views α = 0.90)** are our strongest long-term drivers. A spike in these channels stays in the system nearly forever, suggesting these users are highly embedded.
  - Channels like **Paid Views α = 0.05)** show zero memory and exhibit over 40% zero-value weeks in the data. They act as tactical triggers that only work while being actively "pumped".
- **The Seasonality Effect:**
  - Approximately **15% of Q4 success** is driven by organic holiday demand (November seasonality).
  - **Insight:** The model isolates this baseline, allowing for budget planning that is net of seasonality rather than over-attributing holiday demand to media brilliance.

Strategic Recommendations

- **Prioritize Google and Facebook:** These channels show the best intersection of sales correlation, sheer scale, and healthy decay **(α = 0.70-0.75).** They should remain the primary focus for major marketing efforts.



- **Reallocate Tactical Spend**: Relegate **Paid Views** to a need-by-need basis for short-term spikes. They will not build the brand baseline or provide a stable long-term sales floor.
- **Leverage Affiliate Longevity**: Encourage and expand **Affiliate Partnerships** to take advantage of their exceptionally long decay and anchor effect on sales.
- **Adopt Seasonality-Aware Planning**: Use the dashboard's month dummies to adjust Q4 spend levels, preventing over-spending during peak months where organic demand is already doing the heavy lifting.

Technical Integrity

- **Predictive Strength**: The **0.78 R^2** score provides high confidence that the dashboard's outputs are grounded in real historical patterns

- **Observational Guardrail**: The model identifies associations based on historical data. To confirm causal impact, I recommend running a 4-week geo-lift test in a high-performing division before implementing major budget reallocations.